

B. Tech Degree V Semester Examination, November 2006**ME 505 HEAT AND MASS TRANSFER***(2002 Admissions Onwards)*

Time : 3 Hours

Maximum Marks : 100

(Approved data book may be permitted)

- I. (a) Derive the general heat conduction equation in Cartesian coordinates. (10)
 (b) A steam pipe of 160mm inside diameter and 5mm thick ($K=58 \text{ W/m}^0\text{C}$) is covered with first layer of insulating material 30mm thick ($K=0.17 \text{ W/m}^0\text{C}$) and second layer of insulating material 50mm thick ($K=0.093 \text{ W/m}^0\text{C}$). The temperature of steam passing through the pipe is 300^0C and the ambient air temperature surrounding the pipe is 30^0C . Taking the inner and outer heat transfer coefficients as 30 and $K=5.8 \text{ W/m}^0\text{C}$ respectively, find the heat lost per metre length of the pipe. (10)
- OR**
- II. (a) Explain (i) Critical insulation thickness
 (ii) Fin effectiveness
 (iii) Relaxation method (12)
 (b) A thermocouple junction, which may be approximated as a sphere, is to be used for temperature measurement in a gas stream. The convection coefficient between the junction surface and gas is known to be $400 \text{ W/m}^2\text{K}$, and the junction thermo physical properties are $K=20 \text{ W/mK}$, $C=400 \text{ J/KgK}$ and $\rho = 8500 \text{ Kg/m}^3$. Determine the junction diameter needed for the thermocouple to have a time constant of 1S. If the junction is at 25^0C and is placed in a gas stream that is at 200^0C , how long will it take for the junction to reach 199^0C . (8)
- III. (a) Describe the concept of velocity boundary layer and thermal boundary layer. (8)
 (b) Air at a pressure of 6 KN/m^2 and a temperature of 300^0C flows with a velocity of 10 m/s over a flat plate 0.5 m long. Estimate the cooling rate per unit width of the plate needed to maintain it at a surface temperature of 27^0C . (12)
- OR**
- IV. (a) Explain Reynold's analogy. (5)
 (b) Differentiate between film wise and drop wise condensation and mention their relative merits. (7)
 (c) A vertical pipe 5 cm diameter carrying hot water is exposed to ambient air at 15^0C . If the outer surface of the pipe is 65^0C , find the heat loss from one metre height of the pipe per hour. (8)
- V. (a) Explain the following laws of radiation
 (i) Stefan – Boltzmann law
 (ii) Wien's displacement law (10)
 (b) A glass plate 30 cm square is used to view radiation from a furnace. The transmissivity of the glass is 0.5 from 0.2 to $3.5 \mu \text{ m}$. The emissivity may be assumed to be 0.3 upto 3.5μ and 0.9 above that the transmissivity of the glass is zero, except in the range from 0.2 to 3.5μ . Assuming that the furnace is a black body at 2000^0C , calculate the energy absorbed in the glass and the energy transmitted. (10)

OR*(Turn Over)*

- I. (a) Explain: (i) emissive power
(ii) black body
(iii) radiation shape factor (9)
- (b) Calculate the net radiant heat exchange per m^2 area of two large parallel plates at temperatures of 327°C and 27°C respectively. ϵ (hot plate) = 0.9 and ϵ (cold plate) = 0.6. If a polished aluminium shield is placed between them, find the percentage reduction in the heat transfer ϵ (shield) = 0.4. (11)

- II. (a) Derive an expression for the logarithmic mean temperature difference (LMTD) of a parallel-flow heat exchanger in standard form. (10)
- (b) A counter flow double pipe heat exchanger using superheated steam is used to heat water at the rate of $10,500\text{Kg/hr}$. The steam enters the heat exchanger at 180°C and leaves at 130°C . The inlet and exit temperature of water are 30°C and 80°C respectively. If overall heat transfer coefficient from steam to water is $814\text{W/m}^2\text{C}$, calculate the heat transfer area. (10)

OR

- III. (a) Describe the different types of heat exchangers. (8)
- (b) A hot fluid at 200°C enters a heat exchanger at a mass flow rate of 10^4Kg/h . Its specific heat is 2000J/KgK . It is to be cooled by another fluid entering at 25°C with a mass flow rate of 2500Kg/h and specific heat 400J/Kg K . The overall heat transfer coefficient based on outside area of 20m^2 is $250\text{W/m}^2\text{K}$. Find the exit temperature of the hot fluid when the fluids are in parallel flow. (12)

- IV. (a) Define the following terms used in mass transfer studies
(i) Mass concentration (ii) Molar concentration
(iii) Mass fraction (iv) Mole fraction
(v) Mass flux (vi) Diffusion coefficient (12)
- (b) Oxygen gas at 25°C and a pressure of 2 bar is flowing through a rubber pipe of inside diameter 25mm and wall thickness 2.5mm. The diffusivity of oxygen through rubber is $0.21 \times 10^{-9}\text{m}^2/\text{s}$ and the solubility of oxygen in rubber is $0.21 \times 10^{-3}\text{kmol/m}^3\text{bar}$. Find the loss of oxygen by diffusion per metre length of pipe. (8)

OR

- (a) Explain Fick's law of diffusion. (6)
- (b) Discuss the dependence of mass diffusivity of gases on temperature and pressure. (6)
- (c) Air at 25°C and atmospheric pressure flows with a velocity of 3m/s inside a 10mm diameter tube of 1m length. The inside surface of the tube contains a deposit of naphthalene. Determine the average mass transfer coefficient for the transfer of naphthalene from the pipe surface into the air. For naphthalene air, take $D_{AB} = 0.62 \times 10^{-5}\text{m}^2/\text{s}$. (8)

